



PROCEDURE FOR BUILDING A FLOOR FOR AN UNDERGROUND DIAMOND DRILL

OBJECTIVE

The objective of this procedure is to define safe and efficient methods for building a floor for an underground diamond drill

APPLICATION

This procedure applies to all MACHINES ROGER INTERNATIONAL employees.

RESPONSIBILITIES

The worker is responsible for applying this procedure and the supervisor is responsible for ensuring that operations run smoothly and that the work is done properly.

ALWAYS CARRY LARGE PIECES OF WOOD WITH TWO PEOPLE.

REQUIRED MATERIALS	REQUIRED TOOLS
-Work-in-progress barricade with orange flasher	Crowbar (nail puller)
-Cabinet (First aid kit and eyewash 2x)	Sledgehammer
-Fire extinguisher	Hammer
-Planed 2"X 4" construction lumber	Bucksaw
-Rough 6"X 6" lumber	Log tongs
-Rough 3"X 8" lumber	Bosch laser
-Wooden wedges	48" level
-6" lag screws	Scaling bars 6' 8' and 10'
-Wood blade for saw	Muck scoop
	"Sawzall" with wood blade

Required PPE

Mandatory basic equipment

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PROCEDURE

The supervisor must first have inspected the area before moving the equipment to the drilling location. He must ensure that:

- The floor is free of debris
- The floor is flat
- The “bootlegs” have been located and marked

- 1- With the drilling plan in hand, determine the exact positioning of the future floor based on the planned position of the drill according to the drilling plan, and mark the position of the future floor with paint to guide yourself with the drift so you can plan a straight construction over its full 16’X32’ length



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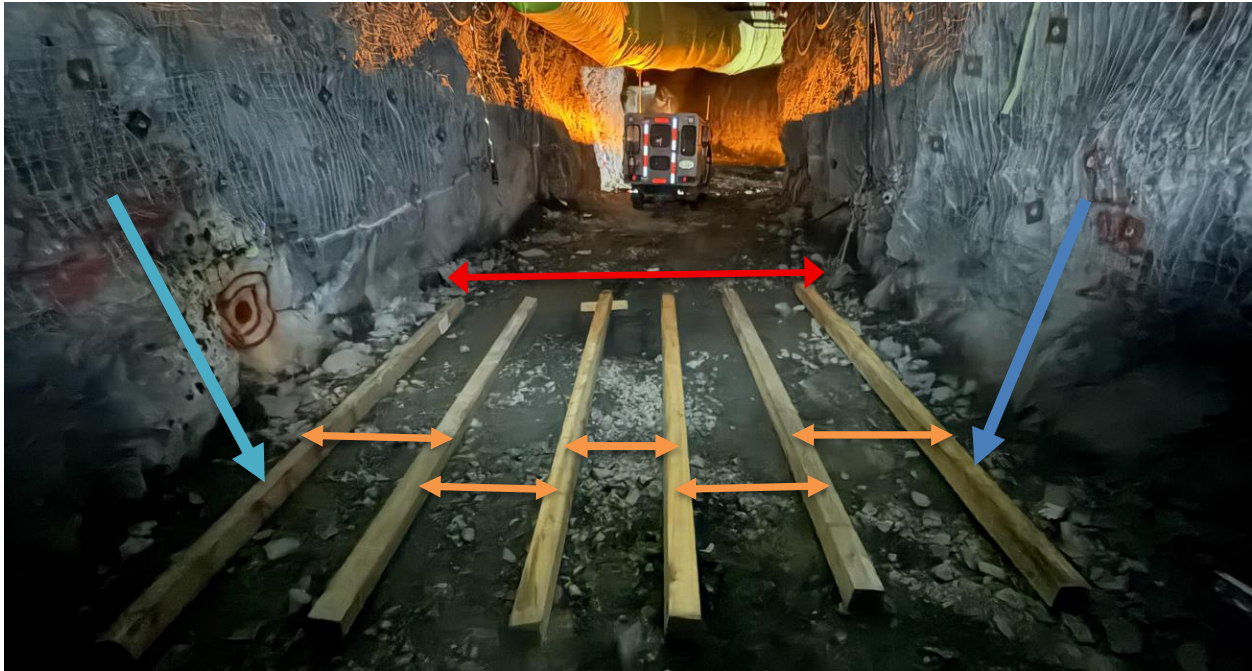
- 2- Determine the highest point of the area where the floor will be built. Use the Bosch laser level to do this. Place your first plank directly on the ground at that location. This plank will serve as the reference point for leveling the other planks of your floor.



- 3- Install the 6"X6" **base timbers** perpendicular to the drilling direction so that the drill skids cross over them, in accordance with the plans and specifications (drilling azimuth), unless otherwise instructed by the supervisor.

****This step is important because it prevents the floor from being raised unnecessarily, which would require more blocking material and additional wood handling, resulting in a considerable loss of time..***

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- 4- Place your first plank on the painted reference line from step (1), then position a 6"X6" at 16' from your first to get a straight 16'X16' frame. Then position the other four 6"X6" at a distance of 32" "center" each
- 5- Level the floor using a 48" level and a piece of 3"X8" at least 8 feet long for greater reach and better results



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- To solidly support your structure and keep it level, cut 3"X8" and/or 6"X6" pieces to make blocking at the desired height.
- You must position the blocking and the wooden wedges perpendicular to the planks, and you must place a minimum of 6 blockings to support each 6"X6"

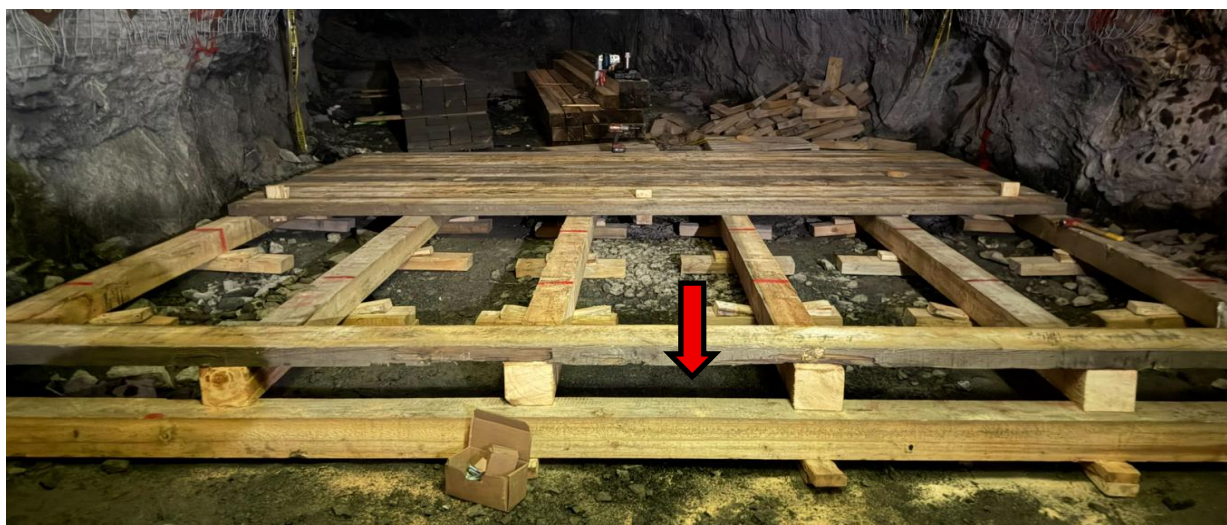


- Use wooden wedges to level the base structure and make it easier to adjust the level of each timber.

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- When you level the end of the 6"X6" of your first 16'X16' section (the one that will end up in the centre of your floor) **place your blocking so it can support the 6"X6" of the next section**

**This method will save you time and ensure you have 2 perfectly equal floor sections.*



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- 6- Once the base is leveled and completed, install the rough 3"X8" lumber, laying it perpendicular to the 6"X6" to cover the floor surface, starting from the beginning and screwing the first plank with **Lag screws** at each end, working towards the end. For safety reasons You must always start covering the floor in the direction where you can walk on the covered surface of 3"X8", so you do not have to step over the 6"X6" planks that serves as the base structure.



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7- After covering about 50% of the surface of a 16'x16' section, you can tighten your planks with wooden wedges to make it more solid and prevent gaps from forming over time, as follows:

- * Screw the last 3''X8'' with a lag screw at each end
- * Insert wooden wedges and drive them in to create tension between the planks
- * Then screw the plank in front of the wooden wedges to hold the tension
- * Remove the lag screws from the last 3''X8'' to release the wooden wedges.



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- 8- When the first 16'X16' section is covered, leave a space of about 2 free planks to allow you to start the base structure of your second section, and at the end you will have the space ***to screw with lag screws to join the sections together on each side of your floor***



- 9- Start the second 16'X16' section of the base structure. To determine the required height, use the Bosh laser level and position the laser beam to determine the height required to support the underside of your 6"X6".

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10-Then position a plank 16 feet from your first section; this will give you the height needed to raise your base structure.

**This method will allow you to start your 2nd section more easily, give you an effective reference point and save you unnecessary handling.*



- When the height needed to level your base structure allows it, use 16 feet 6"X6" as supports, positioning them perpendicular. This will give a stable structure and allow you to place your 6"X6" on top to join the first section
- Using your last 6"X6" from your first section and the one positioned as a guide for the second section, you can easily position at 32" center between each and adjust the other 6"X6" to the same height.

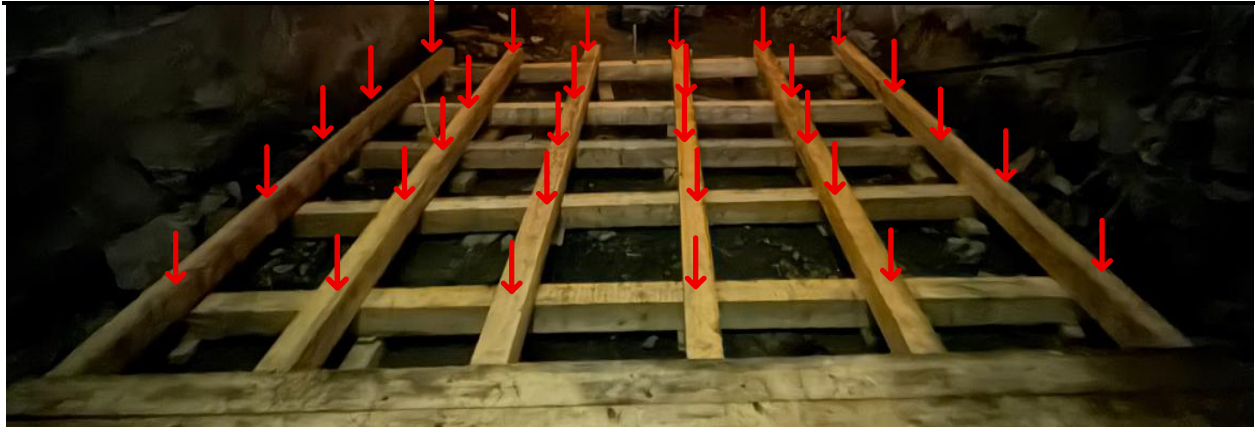
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- After positioning your 6"X6" level **at 32" center**. You will be able to place your others on top to join those of the first section.



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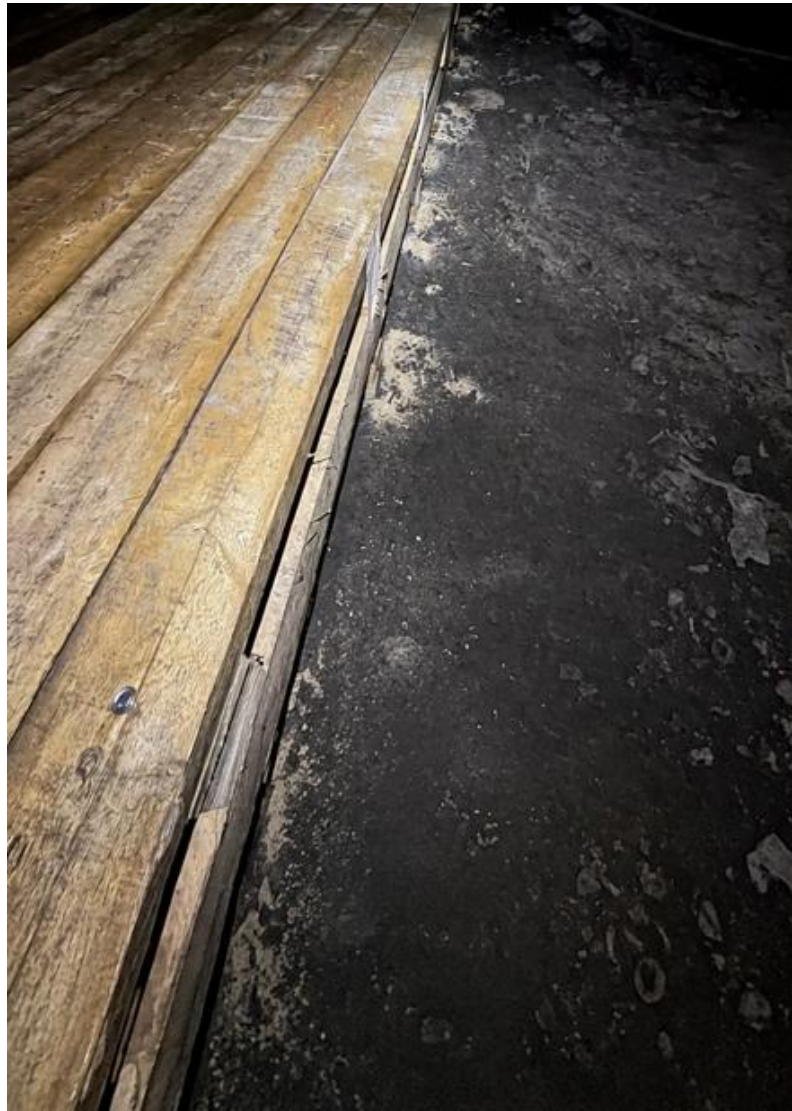
11-Use wooden wedges to even out the base structure and allow you to easily adjust the level of each one. **Always align your supports with the crossing point of your 6"X6"** to ensure solid support.

- In special cases where a crib stack is required to reach the desired height, use 24" to 36" long sections of 3"X8" and/or 6"X6", stacked as shown in the image below.



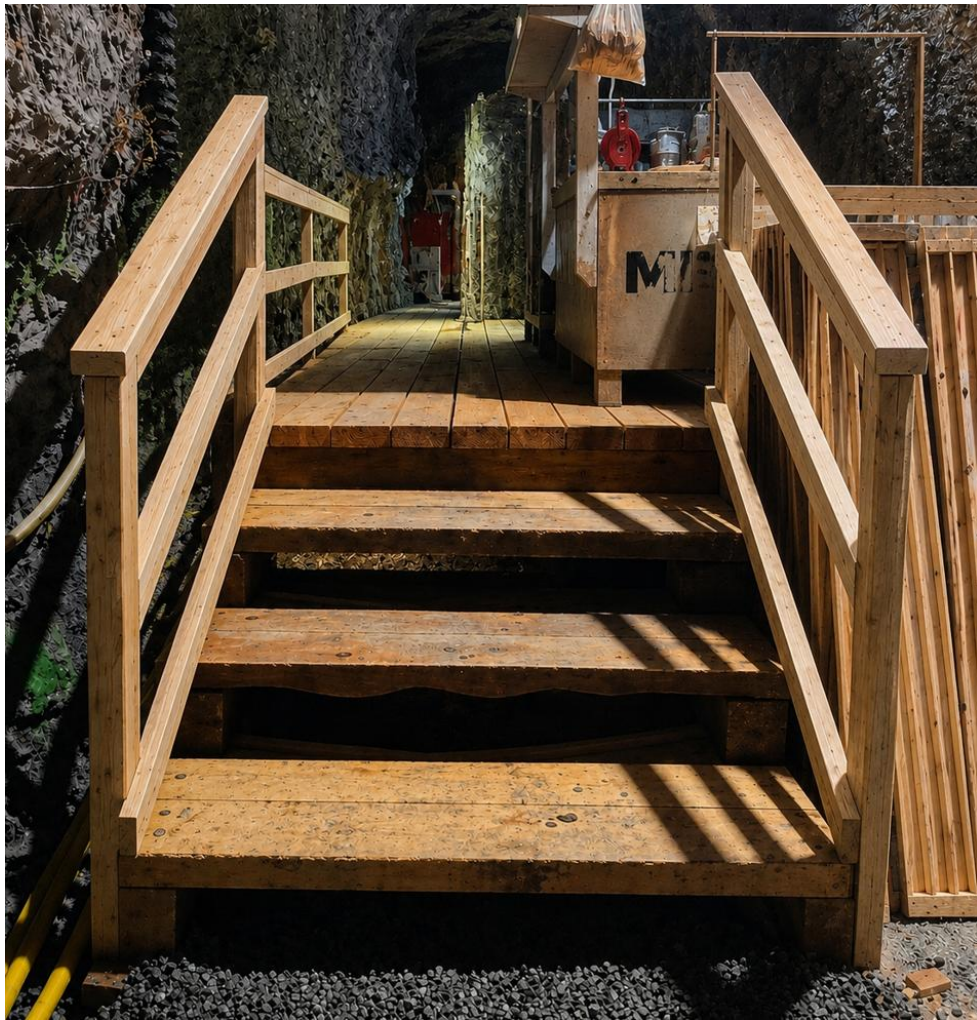
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- 12-Once the base structure of the second floor section is completed, continue covering with 3"X8", *repeating steps 6 and 7 above.*
- 13-When you have completed the entire floor surface, check there is no 6"X6" or 3"X8" pieces of wood sticks out; they must be cut square and flush with the edge of the floor-so as not to trip on uneven pieces.



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- 14- If the floor is higher than 12 inches, steps must be built. They must be at least 16" deep (2 planks of 3"X8"), at least 36" wide, a maximum of 60" wide and a maximum of 12 inches high.



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*Here is an example of a 3-step stacked construction that is simple to build and reusable. Depending on the number of steps needed, they must be adjusted to 9" from the floor and levelled to obtain equal, straight and solid steps.



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If the floor is higher than 12 inches, guardrails must be installed and comply with section 12 of the RSST:

12. Guardrails: Any guardrail incorporated into a building, except for those provided with equipment, must comply with the National Building Code as in effect at the time of installation.

Temporary guardrails must be designed, constructed, and installed to withstand the following minimum loads:

(1) a horizontal point load of 900 N applied at any point on the top rail;

(2) a vertical load of 450 N applied to the top rail.

In addition, temporary guardrails must have:

(1) a top rail located between 1 m and 1.2 m above the floor;

(2) at least one intermediate rail secured at the midpoint between the top rail and the floor. The intermediate rail may be replaced by balusters or panels;

(3) a toe board at floor level at least 90 mm high.

In areas where there is a concentration of workers, as well as in other areas where temporary guardrails may be subjected to extraordinary stresses, they must be reinforced accordingly.



Top rail

Mid rail

Bottom rail
(toe board)

15-A bottom rail must be added at locations on the floor where workers have access.

16- If there is more than one step, handrails will be required.

17-If a gap between your floor and the wall is 12" or less, add a 24" high mesh attached to the bottom of the wall; it will keep you from stepping in and eliminate the risk of injury.

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SECONDARY FLOOR

- 1- If another floor is built to stand higher next to the tower:
 - a. It must be a minimum of 32" wide by 72" long and a maximum of 12" high;
 - b. If the floor requires 2 steps, the first step must be 16 inches deep (2 planks of 3"X8") and 9 inches high; if it has no step on one of its sides, it must have a solidly attached guardrail.
 - c. No piece may overhang, and each piece must be properly supported by crosspieces underneath.
 - d. The last 2 planks closest to the drill must be screwed with lag screws;
 - e. The last plank must come within **at least 3 inches** of the tower;
 - f. **At each change of azimuth, degrees or drilling angle, the last 2 planks must be modified to fit the contour of the drill.**



- g. The guardrails must also be adjusted to close the gap created by the movement
 - h. **Adding planed lumber on the floors is not tolerated**



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DISMANTLING AFTER DRILL DEMOBILIZATION

1. First, dismantle the secondary floor if necessary
2. Then dismantle the main floor, always starting with the planks at the far end of the floor,
3. Always remove the lag screws from the planks when they are removed,
4. When all the planks are removed and all the nails or lag screws pulled, dismantle the 6"X6" base structure.
5. The accumulation of drilling mud under the floor can make moving around difficult and unstable:
 - a. Position yourself in the most solid and stable area;
 - b. Stay vigilant and keep your feet firmly on the ground;
 - c. Start at the far end of the drilling bay;
 - d. Work by sliding the pieces of wood over one another to get them out of position;

AFTER DISMANTLING IS COMPLETE

1. Install an orange fence.
2. Install a red tape card with the following information:
 - a. Hole not surveyed and not cemented
 - b. Installation date
 - c. Supervisor's name



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DECISION CRITERIA FOR CHOOSING FLOOR TYPES BASED ON SPECIAL SITUATIONS

Some situations require floors to be smaller and of different shapes. For example, for short holes with a drill that is not a "MOBILE" type but is used as such.

- 1- **0 to 49 metres** on the same set-up and/or hole without a bit change or rod pulling:
 - a. 10 feet X 8 feet floor without the addition of an upright rods rack (minimum width)
- 2- **If a bit change is known during the drilling of one or more holes on the same set-up and less than 100 m**
 - a. Addition of an upright rods rack (minimum width) depending on the number of rods to pull
- 3- **49 to 499 metres or as soon as we know rods must be pulled for a bit change or other reasons:**
 - a. 10 feet X 16 feet floor with the addition of an upright rod rack (minimum width)
- 4- **500 metres and more:**
 - a. **Full 16 feet X 32 feet floor**

The addition of the rods rack serves to reduce the number of rods to handle and place on the horizontal rack.

Christian St-Amour
 Director of Operations

Corrections made	Date	By
Addition of bottom rail, point 9	19-11-2020	S. Tremblay
Addition of the secondary floor section following an ankle sprain incident	1-09-2022	S. Tremblay
Add OR ROUGH after planed (2X4) in the required materials table due to the lack of availability of planed lumber	27-09-2024	S. Tremblay
Addition of the "Floors in special situations" section	21-01-2025	S. Tremblay S. Desrosiers
Complete revision	27-07-2026	É. Thibeault F. Croteau